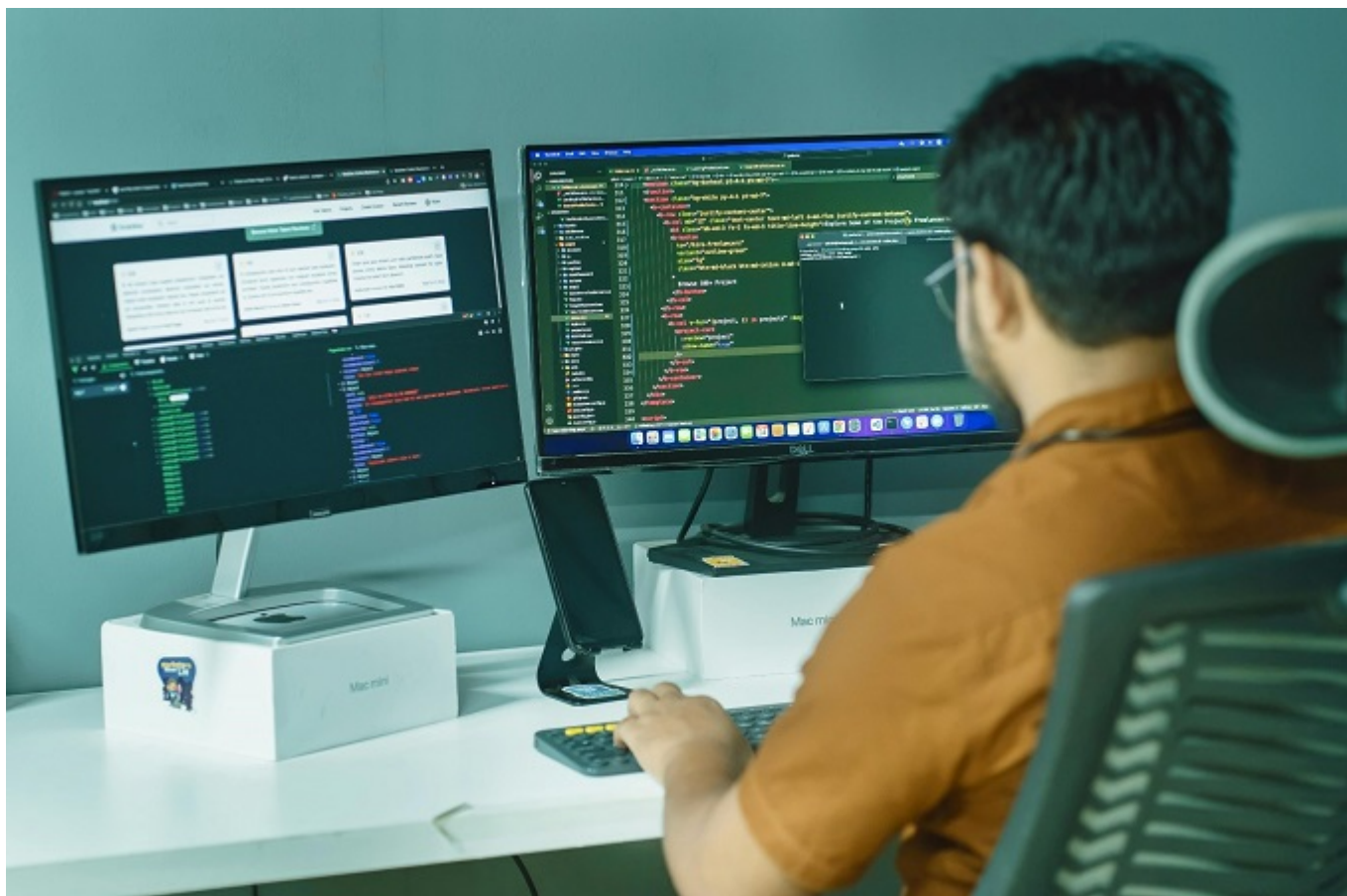


## Qualification Pack



# Web Developer

QP Code: SSC/Q0503

Version: 4.0

NSQF Level: 4

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## Qualification Pack

## Contents

|                                                                                            |    |
|--------------------------------------------------------------------------------------------|----|
| SSC/Q0503: Web Developer .....                                                             | 3  |
| <i>Brief Job Description</i> .....                                                         | 3  |
| Applicable National Occupational Standards (NOS) .....                                     | 3  |
| <i>Compulsory NOS</i> .....                                                                | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                            | 3  |
| SSC/N0501: Contribute to the design of software products and applications .....            | 5  |
| SSC/N0503: Develop Media Content and Graphics for Software Products and Applications ..... | 13 |
| DGT/VSQ/N0101: Employability Skills (30 Hours) .....                                       | 19 |
| Assessment Guidelines and Weightage .....                                                  | 24 |
| <i>Assessment Guidelines</i> .....                                                         | 24 |
| <i>Assessment Weightage</i> .....                                                          | 24 |
| Acronyms .....                                                                             | 26 |
| Glossary .....                                                                             | 28 |

## Qualification Pack

### SSC/Q0503: Web Developer

#### Brief Job Description

Individuals at this job are responsible for designing and maintaining web-based applications that include static and dynamic content. This includes the design, layout and coding of a website. They may work standalone or along with application/functional developers as part of the overall solution that includes a web based component.

#### Personal Attributes

This job may require the individual to work independently or in teams. The individual should be result oriented. The individual should also be able to demonstrate skills for communication, creative and logical thinking.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [SSC/N0501: Contribute to the design of software products and applications](#)
2. [SSC/N0503: Develop Media Content and Graphics for Software Products and Applications](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                                      |                        |
|--------------------------------------|------------------------|
| <b>Sector</b>                        | IT-ITeS                |
| <b>Sub-Sector</b>                    | IT Services            |
| <b>Occupation</b>                    | Application Deployment |
| <b>Country</b>                       | India                  |
| <b>NSQF Level</b>                    | 4                      |
| <b>Credits</b>                       | 12                     |
| <b>Aligned to NCO/ISCO/ISIC Code</b> | NCO-2015/ 2513.0101    |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>12th grade Pass (With Computer Knowledge)<br/>OR<br/>10th grade pass with 3 Years of experience in relevant field. Relevant field: Experience in Programming languages (Such as HTML, CSS, Javascript, etc.) The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications.<br/>OR<br/>Previous relevant Qualification of NSQF Level (3) with 3 Years of experience in relevant field. Relevant field: Experience in Programming languages (Such as HTML, CSS, Javascript, etc.) The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications.</p> |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Pre-Requisite License or Training</b>                  | NIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Next Review Date</b>                                   | 18/02/2028                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>NSQC Approval Date</b>                                 | 18/02/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Version</b>                                            | 4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reference code on NQR</b>                              | QG-04-IT-03650-2025-V2-NASSCOM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>NQR Version</b>                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Qualification Pack

### SSC/N0501: Contribute to the design of software products and applications

#### Description

This unit involves contributing to software design with low complexity and impact, focusing on UI/UX principles, responsive design, front-end and back-end development, version control, and the use of Generative AI for enhanced efficiency and innovation.

#### Scope

The scope covers the following :

- Understanding Business and Software Requirements
- Designing Basic Programming Structures
- Documentation and Compliance
- UI/UX Design Principles
- Responsive and Adaptive Design
- Front-End Development
- Back-End Development
- Version Control and Deployment
- Web Performance Optimization
- Testing and Debugging
- Generative AI

#### Elements and Performance Criteria

##### *Understanding Business and Software Requirements*

To be competent, the user/individual on the job must be able to:

- PC1.** collaborate with project stakeholders to fully comprehend Business Requirements Specification (BRS), User Requirements Specification (URS), and Software Requirements Specification (SRS)
- PC2.** ensure the understanding of High-Level Design (HLD) concept by consulting with relevant stakeholder

##### *Designing Basic Programming Structures*

To be competent, the user/individual on the job must be able to:

- PC3.** design basic programming structures to implement functionality in line with BRS/URS, SRS, and HLD
- PC4.** conduct design review meetings with industry experts and senior developers to validate the design against industry standards and best practices
- PC5.** analyze the inputs to identify, resolve, and record design defects and inform future designs

##### *Documentation and Compliance*

To be competent, the user/individual on the job must be able to:

- PC6.** document the details of designs using standard templates and tools (e.g., UML diagrams, wireframes)

## Qualification Pack

- PC7.** comply with organizational policies, procedures, and guidelines when contributing to the design of software products and applications

### *UI/UX Design Principles*

To be competent, the user/individual on the job must be able to:

- PC8.** apply user-centric design principles by developing wireframes, prototypes, and mockups that meet user expectations
- PC9.** ensure all designs are compliant with accessibility standards, including WCAG (Web Content Accessibility Guidelines)

### *Responsive and Adaptive Design*

To be competent, the user/individual on the job must be able to:

- PC10.** test web applications for cross-browser compatibility to ensure consistent functionality and appearance across different browsers and platforms
- PC11.** apply mobile-first and responsive design practices using frameworks like Bootstrap

### *Front-End Development*

To be competent, the user/individual on the job must be able to:

- PC12.** develop front-end functionality using technologies such as HTML5, CSS3, JavaScript, and frameworks like React or Angular

### *Back-End Development*

To be competent, the user/individual on the job must be able to:

- PC13.** integrate with databases (e.g., MySQL, MongoDB) to support back-end functionality
- PC14.** apply server-side programming with languages like Node.js, PHP, or Python.

### *Version Control and Deployment*

To be competent, the user/individual on the job must be able to:

- PC15.** use Git for version control to manage code and collaboration
- PC16.** implement Continuous Integration/Continuous Deployment (CI/CD) pipelines for smooth deployment

### *Web Performance Optimization*

To be competent, the user/individual on the job must be able to:

- PC17.** implement techniques like lazy loading and code splitting to enhance page loading times
- PC18.** optimize images and resources for better performance

### *Testing and Debugging*

To be competent, the user/individual on the job must be able to:

- PC19.** use tools like Chrome DevTools to debug front-end issues and optimize performance
- PC20.** develop unit and integration tests using frameworks like Jest or Mocha for reliable code

### *Generative AI*

To be competent, the user/individual on the job must be able to:

- PC21.** use Generative AI tools like OpenAI, Google Bard, or Microsoft Copilot to automate tasks such as ticket categorization and prompt responses
- PC22.** utilize Generative AI for documentation, including creating resolutions for incidents and automating repetitive tasks
- PC23.** identify opportunities to improve issue resolution speed and accuracy using Generative AI tools and other innovative solutions

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the standard operating procedures for creating documents and sharing knowledge to ensure effective internal communication.
- KU2.** how to access and update the organization's knowledge base to maintain current and relevant information.
- KU3.** the purpose and scope of tasks while ensuring adherence to project boundaries and avoiding scope creep.
- KU4.** how to receive, analyze, and implement the feedback to refine and improve work output to meet expectations
- KU5.** how to use standard templates and tools to clearly and consistently document designs in line with organizational standards
- KU6.** the approval process for software design documents to align with internal procedures.
- KU7.** how to design the basic program structures, software products, and applications to fulfill business and user requirements
- KU8.** the process of identifying different sources of information to support the design of software products and specifications.
- KU9.** how to recognize the common design defects and resolving them promptly to ensure high-quality outputs.
- KU10.** how to apply the current infrastructure design practices to ensure the scalability, security, and performance of software products.
- KU11.** the use of testing methods to verify that software products function as intended and meet user requirements
- KU12.** the impact of new products and applications on business processes and infrastructure, ensuring alignment with business goals
- KU13.** how to apply the UI/UX design principles, including user-centric design, wireframing, and prototyping, to create intuitive user interfaces
- KU14.** the mobile-first and responsive design principles using frameworks like Bootstrap to ensure compatibility across devices
- KU15.** the front-end technologies such as Html5, CSS3, and JavaScript to develop dynamic, interactive websites and applications
- KU16.** the back-end technologies like MySQL, PHP, and Python to create efficient server-side applications and manage databases
- KU17.** how to manage the version control and deployment processes, including CI/CD pipelines, to automate and streamline software development and delivery

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** . Review pertinent literature to stay informed about the latest advancements in the field
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker

### Qualification Pack

- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understanding Business and Software Requirements</i>                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | <b>1</b>   |
| <b>PC1.</b> collaborate with project stakeholders to fully comprehend Business Requirements Specification (BRS), User Requirements Specification (URS), and Software Requirements Specification (SRS) | 1            | 2               | -             | 0.5        |
| <b>PC2.</b> ensure the understanding of High-Level Design (HLD) concept by consulting with relevant stakeholder                                                                                       | 1            | 2               | -             | 0.5        |
| <i>Designing Basic Programming Structures</i>                                                                                                                                                         | <b>4</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC3.</b> design basic programming structures to implement functionality in line with BRS/URS, SRS, and HLD                                                                                         | 1.5          | 2               | -             | 1          |
| <b>PC4.</b> conduct design review meetings with industry experts and senior developers to validate the design against industry standards and best practices                                           | 1            | 2               | -             | 1          |
| <b>PC5.</b> analyze the inputs to identify, resolve, and record design defects and inform future designs                                                                                              | 1.5          | 2               | -             | 1          |
| <i>Documentation and Compliance</i>                                                                                                                                                                   | <b>2</b>     | <b>2</b>        | -             | <b>1</b>   |
| <b>PC6.</b> document the details of designs using standard templates and tools (e.g., UML diagrams, wireframes)                                                                                       | 1            | 1               | -             | 0.5        |
| <b>PC7.</b> comply with organizational policies, procedures, and guidelines when contributing to the design of software products and applications                                                     | 1            | 1               | -             | 0.5        |
| <i>UI/UX Design Principles</i>                                                                                                                                                                        | <b>4</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC8.</b> apply user-centric design principles by developing wireframes, prototypes, and mockups that meet user expectations                                                                        | 2            | 2.5             | -             | 1          |
| <b>PC9.</b> ensure all designs are compliant with accessibility standards, including WCAG (Web Content Accessibility Guidelines)                                                                      | 2            | 2.5             | -             | 1          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Responsive and Adaptive Design</i>                                                                                                                        | <b>3</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC10.</b> test web applications for cross-browser compatibility to ensure consistent functionality and appearance across different browsers and platforms | 1.5          | 2.5             | -             | 1          |
| <b>PC11.</b> apply mobile-first and responsive design practices using frameworks like Bootstrap                                                              | 1.5          | 2.5             | -             | 1          |
| <i>Front-End Development</i>                                                                                                                                 | <b>1</b>     | <b>3</b>        | -             | <b>1</b>   |
| <b>PC12.</b> develop front-end functionality using technologies such as HTML5, CSS3, JavaScript, and frameworks like React or Angular                        | 1            | 3               | -             | 1          |
| <i>Back-End Development</i>                                                                                                                                  | <b>3</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC13.</b> integrate with databases (e.g., MySQL, MongoDB) to support back-end functionality                                                               | 1.5          | 3               | -             | 1          |
| <b>PC14.</b> apply server-side programming with languages like Node.js, PHP, or Python.                                                                      | 1.5          | 3               | -             | 1          |
| <i>Version Control and Deployment</i>                                                                                                                        | <b>3</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC15.</b> use Git for version control to manage code and collaboration                                                                                    | 1.5          | 2.5             | -             | 1          |
| <b>PC16.</b> implement Continuous Integration/Continuous Deployment (CI/CD) pipelines for smooth deployment                                                  | 1.5          | 2.5             | -             | 1          |
| <i>Web Performance Optimization</i>                                                                                                                          | <b>3</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC17.</b> implement techniques like lazy loading and code splitting to enhance page loading times                                                         | 1.5          | 2.5             | -             | 1          |
| <b>PC18.</b> optimize images and resources for better performance                                                                                            | 1.5          | 2.5             | -             | 1          |
| <i>Testing and Debugging</i>                                                                                                                                 | <b>2</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC19.</b> use tools like Chrome DevTools to debug front-end issues and optimize performance                                                               | 1            | 2.5             | -             | 1          |
| <b>PC20.</b> develop unit and integration tests using frameworks like Jest or Mocha for reliable code                                                        | 1            | 2.5             | -             | 1          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Generative AI</i>                                                                                                                                     | 3            | 4               | -             | 2          |
| <b>PC21.</b> use Generative AI tools like OpenAI, Google Bard, or Microsoft Copilot to automate tasks such as ticket categorization and prompt responses | 1            | 2               | -             | 1          |
| <b>PC22.</b> utilize Generative AI for documentation, including creating resolutions for incidents and automating repetitive tasks                       | 1            | 1               | -             | 0.5        |
| <b>PC23.</b> identify opportunities to improve issue resolution speed and accuracy using Generative AI tools and other innovative solutions              | 1            | 1               | -             | 0.5        |
| <b>NOS Total</b>                                                                                                                                         | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| <b>NOS Code</b>            | SSC/N0501                                                      |
| <b>NOS Name</b>            | Contribute to the design of software products and applications |
| <b>Sector</b>              | IT-ITeS                                                        |
| <b>Sub-Sector</b>          | IT Services                                                    |
| <b>Occupation</b>          | Application Deployment                                         |
| <b>NSQF Level</b>          | 4                                                              |
| <b>Credits</b>             | 4                                                              |
| <b>Version</b>             | 3.0                                                            |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                     |
| <b>Next Review Date</b>    | 18/02/2028                                                     |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                     |

## Qualification Pack

# SSC/N0503: Develop Media Content and Graphics for Software Products and Applications

## Description

This unit is about developing media content and producing graphic designs to enhance to usability of software products and applications where their business impact and technical complexity is low.

## Scope

The scope covers the following :

- Media Content and Graphic Design Development
- Responsive Web Design
- Web Accessibility Standards
- Dynamic Content Integration
- Review and Feedback Process
- Documentation and Knowledge Sharing
- Compliance and Approval

## Elements and Performance Criteria

### *Media Content and Graphic Design Development*

To be competent, the user/individual on the job must be able to:

- PC1.** convert Business Requirements Specification (BRS) and Software Requirements Specification (SRS) into media content and graphic designs, using available resources and tools
- PC2.** access reusable components, media, and graphical packages from the organization's knowledge base to optimize design work
- PC3.** utilize reusable components to create dynamic and responsive media content and graphic designs
- PC4.** ensure the media content and graphic designs align with the overall design specifications and business objectives
- PC5.** collaborate with stakeholders to gather feedback and refine media content and graphic designs to meet user needs
- PC6.** maintain consistency in design elements, ensuring alignment with brand guidelines and user interface standards

### *Responsive Web Design*

To be competent, the user/individual on the job must be able to:

- PC7.** design media content and graphics optimized for various screen sizes and devices, ensuring responsiveness and fluid layout across platforms
- PC8.** apply mobile-first design principles and use frameworks like Bootstrap to create adaptable web designs

### *Web Accessibility Standards*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

**PC9.** evaluate and test media content and graphic designs for compliance with web accessibility standards (WCAG), ensuring usability for all users

**PC10.** collaborate with accessibility experts to address potential barriers in media content and graphic designs, making necessary adjustments for full inclusivity

### *Dynamic Content Integration*

To be competent, the user/individual on the job must be able to:

**PC11.** integrate dynamic content elements, such as interactive graphics or data-driven media, into software applications to enhance user experience and engagement

### *Review and Feedback Process*

To be competent, the user/individual on the job must be able to:

**PC12.** assess media content and graphic designs with superiors to measure quality and effectiveness, incorporating their input for improvements

**PC13.** modify media content and graphic designs based on feedback, ensuring they meet project objectives and specifications

### *Documentation and Knowledge Sharing*

To be competent, the user/individual on the job must be able to:

**PC14.** record defects and corrective actions taken during the design process to inform future work and design improvements

**PC15.** document insights and experiences gained from developing media content and graphic designs, and update the organization's knowledge base accordingly

### *Compliance and Approval*

To be competent, the user/individual on the job must be able to:

**PC16.** present the finalized media content and graphic designs to supervisors for approval, ensuring adherence to all relevant guidelines and standards.

**PC17.** comply with organizational policies, procedures, and guidelines when creating media content and graphic designs.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** the scope of work to be executed and the importance of staying within ones competency and authority levels to ensure accountability

**KU2.** the standard templates and tools for documenting media content and graphic designs, facilitating consistency across projects.

**KU3.** how to analyze and incorporate feedback from stakeholders to improve media content and graphic designs for better user experience.

**KU4.** the key stakeholders who provide valuable feedback on media content and graphic designs, including superiors and project managers.

**KU5.** the organizations approval process for media content and graphic designs, ensuring compliance with internal standards.

**KU6.** how to convert design specifications into effective media content and graphic designs that align with project goals.

## Qualification Pack

- KU7.** the current practices in developing media content and graphic designs, including industry trends and emerging technologies.
- KU8.** how recording corrective actions for defects can enhance future designs and streamline the development process
- KU9.** how to create, review, and execute media content and graphic designs that meet project objectives and user needs
- KU10.** the criteria for assessing the suitability of components for re-use, promoting efficiency in the design process.
- KU11.** the common issues and defects that may arise in media content development and how to resolve them using best practices
- KU12.** how to test media content and graphic designs to ensure they meet their intended purpose and deliver expected outcomes
- KU13.** the different sources of information to aid in the development of media content and graphic designs, including reference materials and tools.
- KU14.** the principles of responsive web design, ensuring that designs provide optimal user experiences across devices and screen sizes.
- KU15.** the adherence to web accessibility standards (e.g., WCAG), ensuring inclusivity for all users, regardless of disabilities
- KU16.** the methods for integrating dynamic content into web designs to enhance interactivity and user engagement, keeping the design fresh and engaging

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** Read the relevant literature to stay informed about the latest developments in your field.
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Media Content and Graphic Design Development</i>                                                                                                                                     | <b>12</b>    | <b>18</b>       | -             | <b>6</b>   |
| <b>PC1.</b> convert Business Requirements Specification (BRS) and Software Requirements Specification (SRS) into media content and graphic designs, using available resources and tools | 2            | 3               | -             | 1          |
| <b>PC2.</b> access reusable components, media, and graphical packages from the organization's knowledge base to optimize design work                                                    | 2            | 3               | -             | 1          |
| <b>PC3.</b> utilize reusable components to create dynamic and responsive media content and graphic designs                                                                              | 2            | 3               | -             | 1          |
| <b>PC4.</b> ensure the media content and graphic designs align with the overall design specifications and business objectives                                                           | 2            | 3               | -             | 1          |
| <b>PC5.</b> collaborate with stakeholders to gather feedback and refine media content and graphic designs to meet user needs                                                            | 2            | 3               | -             | 1          |
| <b>PC6.</b> maintain consistency in design elements, ensuring alignment with brand guidelines and user interface standards                                                              | 2            | 3               | -             | 1          |
| <i>Responsive Web Design</i>                                                                                                                                                            | <b>4</b>     | <b>8</b>        | -             | <b>3</b>   |
| <b>PC7.</b> design media content and graphics optimized for various screen sizes and devices, ensuring responsiveness and fluid layout across platforms                                 | 2            | 4               | -             | 1.5        |
| <b>PC8.</b> apply mobile-first design principles and use frameworks like Bootstrap to create adaptable web designs                                                                      | 2            | 4               | -             | 1.5        |
| <i>Web Accessibility Standards</i>                                                                                                                                                      | <b>4</b>     | <b>7</b>        | -             | <b>3</b>   |
| <b>PC9.</b> evaluate and test media content and graphic designs for compliance with web accessibility standards (WCAG), ensuring usability for all users                                | 2            | 3.5             | -             | 1.5        |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> collaborate with accessibility experts to address potential barriers in media content and graphic designs, making necessary adjustments for full inclusivity | 2            | 3.5             | -             | 1.5        |
| <i>Dynamic Content Integration</i>                                                                                                                                        | <b>2</b>     | <b>3</b>        | -             | <b>2</b>   |
| <b>PC11.</b> integrate dynamic content elements, such as interactive graphics or data-driven media, into software applications to enhance user experience and engagement  | 2            | 3               | -             | 2          |
| <i>Review and Feedback Process</i>                                                                                                                                        | <b>4</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC12.</b> assess media content and graphic designs with superiors to measure quality and effectiveness, incorporating their input for improvements                     | 2            | 3               | -             | 1          |
| <b>PC13.</b> modify media content and graphic designs based on feedback, ensuring they meet project objectives and specifications                                         | 2            | 3               | -             | 1          |
| <i>Documentation and Knowledge Sharing</i>                                                                                                                                | <b>2</b>     | <b>4</b>        | -             | <b>2</b>   |
| <b>PC14.</b> record defects and corrective actions taken during the design process to inform future work and design improvements                                          | 1            | 2               | -             | 1          |
| <b>PC15.</b> document insights and experiences gained from developing media content and graphic designs, and update the organization's knowledge base accordingly         | 1            | 2               | -             | 1          |
| <i>Compliance and Approval</i>                                                                                                                                            | <b>2</b>     | <b>4</b>        | -             | <b>2</b>   |
| <b>PC16.</b> present the finalized media content and graphic designs to supervisors for approval, ensuring adherence to all relevant guidelines and standards.            | 1            | 2               | -             | 1          |
| <b>PC17.</b> comply with organizational policies, procedures, and guidelines when creating media content and graphic designs.                                             | 1            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                          | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                           |
|----------------------------|---------------------------------------------------------------------------|
| <b>NOS Code</b>            | SSC/N0503                                                                 |
| <b>NOS Name</b>            | Develop Media Content and Graphics for Software Products and Applications |
| <b>Sector</b>              | IT-ITeS                                                                   |
| <b>Sub-Sector</b>          | IT Services                                                               |
| <b>Occupation</b>          | Application Deployment                                                    |
| <b>NSQF Level</b>          | 4                                                                         |
| <b>Credits</b>             | 7                                                                         |
| <b>Version</b>             | 3.0                                                                       |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                                |
| <b>Next Review Date</b>    | 18/02/2028                                                                |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                                |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 18/02/2025                      |
| <b>Next Review Date</b>    | 18/02/2028                      |
| <b>NSQC Clearance Date</b> | 18/02/2025                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

The assessment shall be conducted through an online proctored format, incorporating scenario-based multiple-choice questions designed to effectively evaluate practical understanding and real-world application of concepts. Additionally, it will include a viva-voice and hands-on practical evaluation to comprehensively assess the individual's proficiency in specific learning outcomes.

### Minimum Aggregate Passing % at QP Level : 70

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS



**Qualification Pack**

| National Occupational Standards                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| SSC/N0501.Contribute to the design of software products and applications            | 30           | 50              | 0             | 20         | 100         | 45         |
| SSC/N0503.Develop Media Content and Graphics for Software Products and Applications | 30           | 50              | 0             | 20         | 100         | 45         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                                       | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                                        | <b>80</b>    | <b>130</b>      | <b>-</b>      | <b>40</b>  | <b>250</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| <b>NOS</b>     | National Occupational Standard(s)                                |
| <b>NSQF</b>    | National Skills Qualifications Framework                         |
| <b>QP</b>      | Qualifications Pack                                              |
| <b>TVET</b>    | Technical and Vocational Education and Training                  |
| <b>IT-ITeS</b> | Information Technology - Information Technology enabled Services |
| <b>BPM</b>     | Business Process Management                                      |
| <b>BPO</b>     | Business Process Outsourcing                                     |
| <b>KPO</b>     | Knowledge Process Outsourcing                                    |
| <b>LPO</b>     | Legal Process Outsourcing                                        |
| <b>IPO</b>     | Information Process Outsourcing                                  |
| <b>IT-ITeS</b> | Information Technology - Information Technology enabled Services |
| <b>BPM</b>     | Business Process Management                                      |
| <b>BPO</b>     | Business Process Outsourcing                                     |
| <b>KPO</b>     | Knowledge Process Outsourcing                                    |
| <b>LPO</b>     | Legal Process Outsourcing                                        |
| <b>IPO</b>     | Information Process Outsourcing                                  |
| <b>IT-ITeS</b> | Information Technology - Information Technology enabled Services |
| <b>BPM</b>     | Business Process Management                                      |
| <b>BPO</b>     | Business Process Outsourcing                                     |
| <b>KPO</b>     | Knowledge Process Outsourcing                                    |
| <b>LPO</b>     | Legal Process Outsourcing                                        |
| <b>IPO</b>     | Information Process Outsourcing                                  |
| <b>IT-ITeS</b> | Information Technology - Information Technology enabled Services |
| <b>BPM</b>     | Business Process Management                                      |
| <b>BPO</b>     | Business Process Outsourcing                                     |

### Qualification Pack

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| <b>KPO</b>     | Knowledge Process Outsourcing                                    |
| <b>LPO</b>     | Legal Process Outsourcing                                        |
| <b>IPO</b>     | Information Process Outsourcing                                  |
| <b>IT-ITeS</b> | Information Technology - Information Technology enabled Services |
| <b>BPM</b>     | Business Process Management                                      |
| <b>BPO</b>     | Business Process Outsourcing                                     |
| <b>KPO</b>     | Knowledge Process Outsourcing                                    |
| <b>LPO</b>     | Legal Process Outsourcing                                        |
| <b>IPO</b>     | Information Process Outsourcing                                  |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |
| <b>Helpdesk</b>                         | Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.                                                                                                                                                                                                                    |
| <b>Helpdesk</b>                         | Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.                                                                                                                                                                                                                    |
| <b>Helpdesk</b>                         | Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.                                                                                                                                                                                                                    |
| <b>Helpdesk</b>                         | Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.                                                                                                                                                                                                                    |
| <b>Helpdesk</b>                         | Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.                                                                                                                                                                                                                    |